MASTER APPLIED DATA SCIENCE

BAYES INFERENCE

12 OCTOBER 2026

HAN_UNIVERSITY
OF APPLIED SCIENCES

LEARNING GOALS

- The student can explain the Frequentist and Bayesian approach to statistics and knows their main differences and application fields

- The student can explain what the naïve Bayesian technique entails and how it can be used to make predictions on data.

CONTENT

1. The Bayesian approach to statistics
2. Bayes Theorem
3. Bayesian parameter estimation
4. Choosing a prior
5. Frequentist vs. Bayesian approach to statistics
6. Naïve Bayes Classifiers

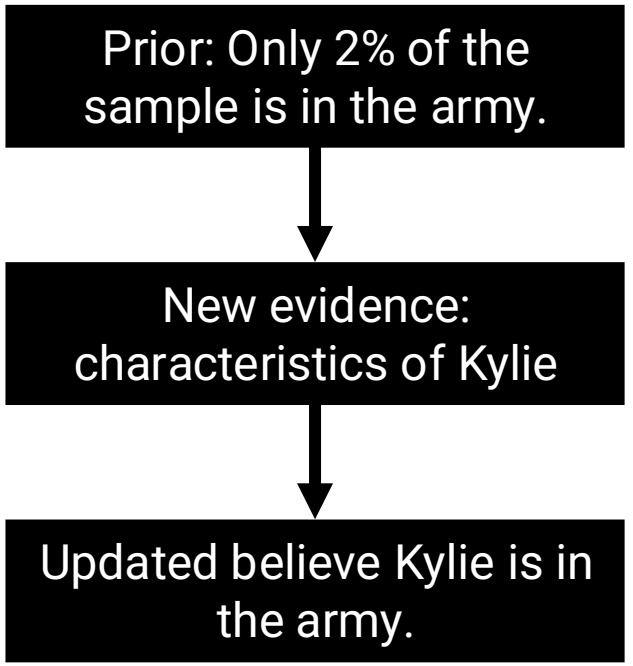
THOUGHT EXPERIMENT

In the Netherlands we have +- 6.000 women in the army.

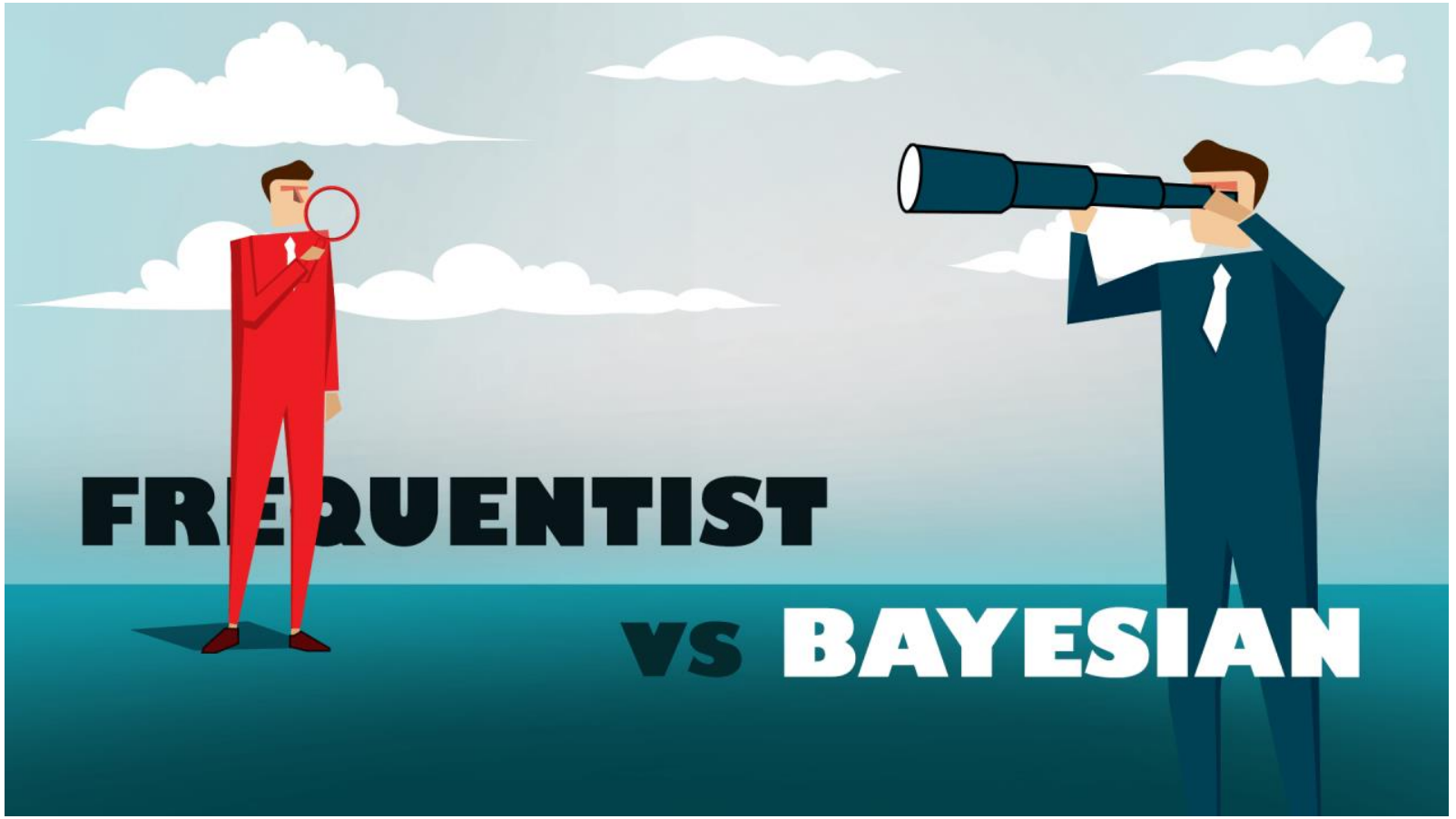
In the Netherlands we have +- 211.000 female nurses.

So, in a random sample of women in the army and female nurses, only 2% will be in the army!

Don’t forget your prior…



How likely is the data under our hypothesis?



What do we believe given the data?

PROBABILITIES

Reliable probabilities require many repeated observations

But in practice:
- Events may be **rare or unique**
- Data may be **scarce**
- We often have relevant **prior knowledge**

How should we reason about uncertainty?



1. THE BAYESIAN APPROACH TO STATISTICS

Suppose a doctor is testing for a rare disease. The disease has a prevalence of 0.01% (1 in 10.000) in the population, and the test has a sensitivity of 99% and specificity of 95%. How likely is the patient to have the disease if the test comes back positive?

- P(Disease) = 0.0001
- (Positive Test | Disease) = 0.99
- (Negative Test | No disease) = 0.95

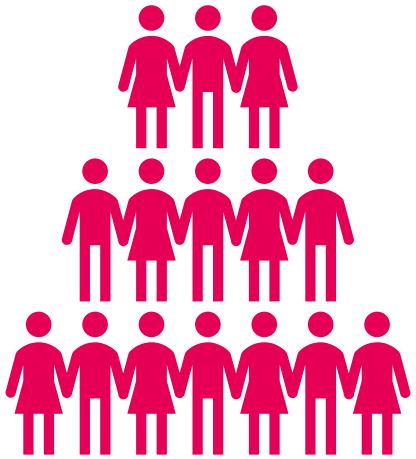
- P(disease | positive test) =
  p(disease) * p(positive test | disease) / p(positive test)



(0.0001 * 0.99) / (0.999*0.05) + (0.99*0.0001) = 0.002 = 0.2%

1. THE BAYESIAN APPROACH TO STATISTICS

Now the doctor gets back the test results and sees according to the positive results the probability the patient has the disease is approximately 0.2%. This is very confusing, the test has high specificity and sensitivity, why would I even test?



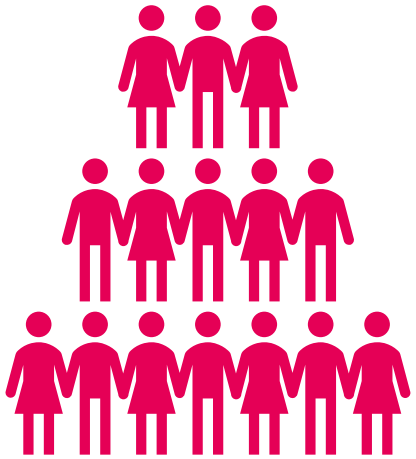
Frequentist

Start with population-level properties and apply them to individuals.



1. THE BAYESIAN APPROACH TO STATISTICS

Due to other symptoms the doctor believes the probability that the patient has the disease (prior to testing) is closer to 30%. How can this previous knowledge be incorporated in the estimate?



Bayesian

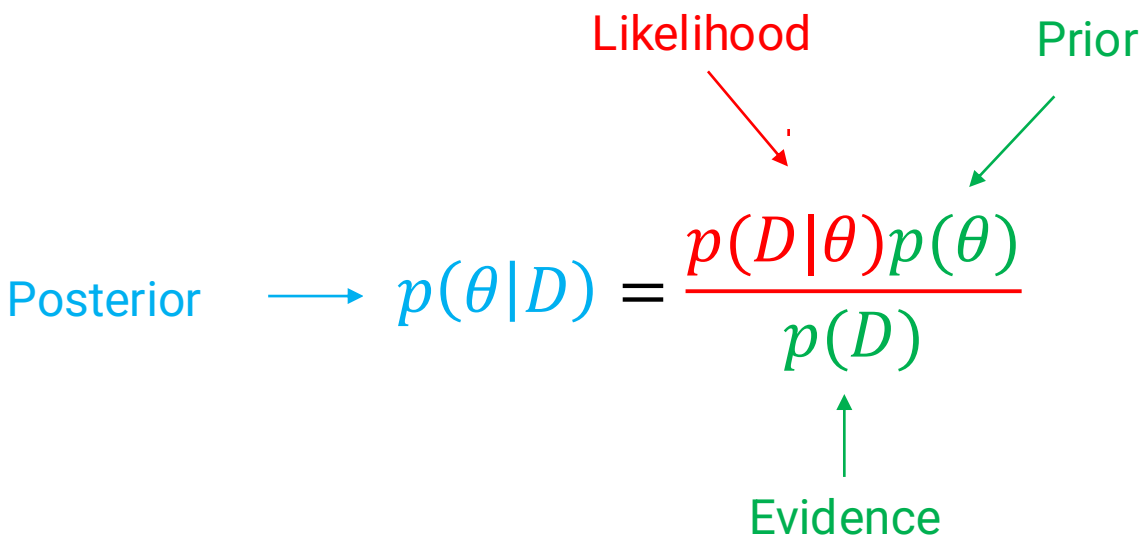
Start from the patient's prior risk and update it using population data (test characteristics).



2. BAYES THEOREM

Bayes theorem provides a method to update probabilities based on experience and data.

It essentially answers the question: Given our current believes (prior) and the data (evidence) how do we update our believes (posterior)?



p(disease) = ~~0.001~~ = 0.3
p(positive test | disease) = 0.99
p(negative test | no disease) = 0.95

2. BAYES THEOREM

Due to other symptoms the doctor beliefs the probability that the patient has the disease (prior to testing) is closer to 30%. How can this belief be incorporated in the estimate?

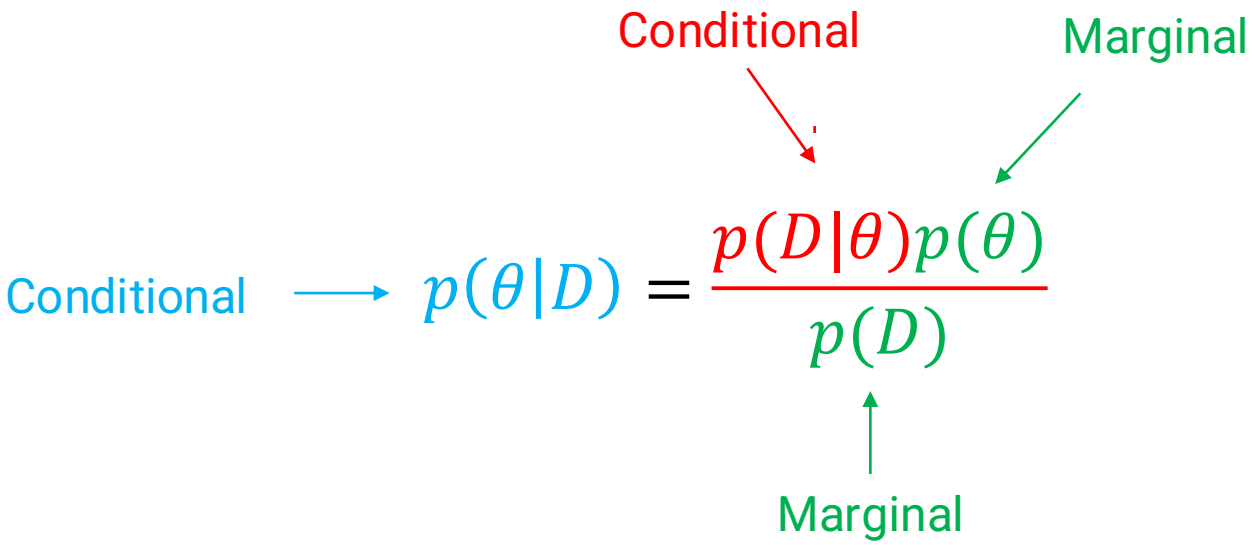
$$p(disease|positive) = \frac{p(positive\ test|disease)p(disease)}{p(positive\ test)}$$

(0.3 * 0.999) / (0.7 * 0.05) + (0.999* 0.3) = 0.89 = 89%

2. BAYES THEOREM

Bayes theorem provides a method to update probabilities based on experience and data.

It essentially answers the question: Given our current believes (prior) and the data (evidence) how do we update our believes (posterior)?



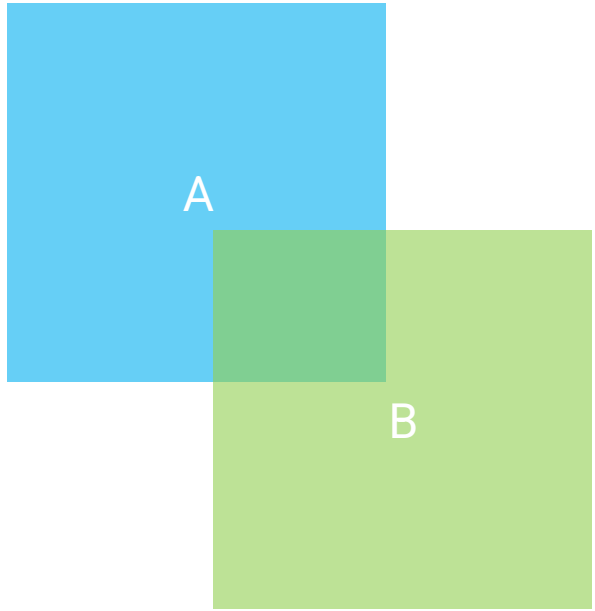
2. BAYES THEOREM

Conditional probability of event A given event B is the joint probability of event A and B divided by the probability of event B (normalizing).

Joint probability

$$p(A|B) = \frac{p(A,B)}{p(B)} = \frac{p(B|A)p(A)}{p(B)}$$

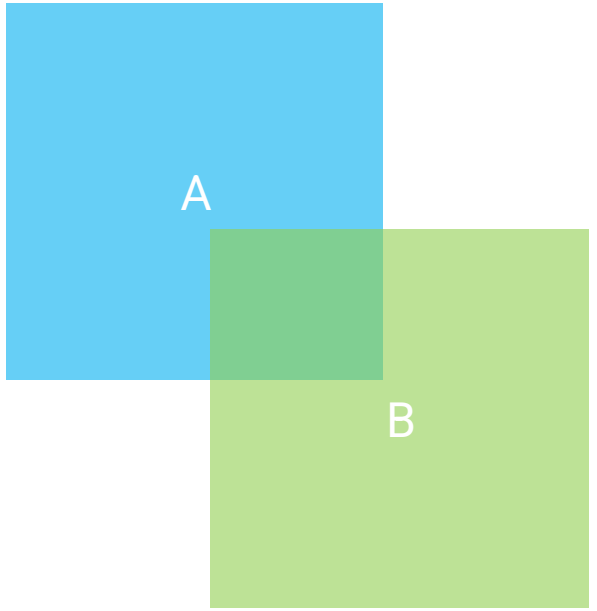
Marginal probability



2. BAYESIAN THEOREM

Marginal: The probability of B summed over all possible values of A.

$$p(B) = \sum_A p(B, A) \qquad p(B) = \sum_A p(B|A)p(A)$$



Suppose B and A are binary, what is p(B)?

- p(B|A=1) = 0.4
- p(B|A=0) = 0.3
- p(A) = 0.5

3. BAYESIAN PARAMETER ESTIMATION

Suppose your friend selects a random coin from a pool of coins with each coin having a probability of heads between 0 and 1.

- What probability of heads would you guess the coin has?
- How certain are you about your guess?



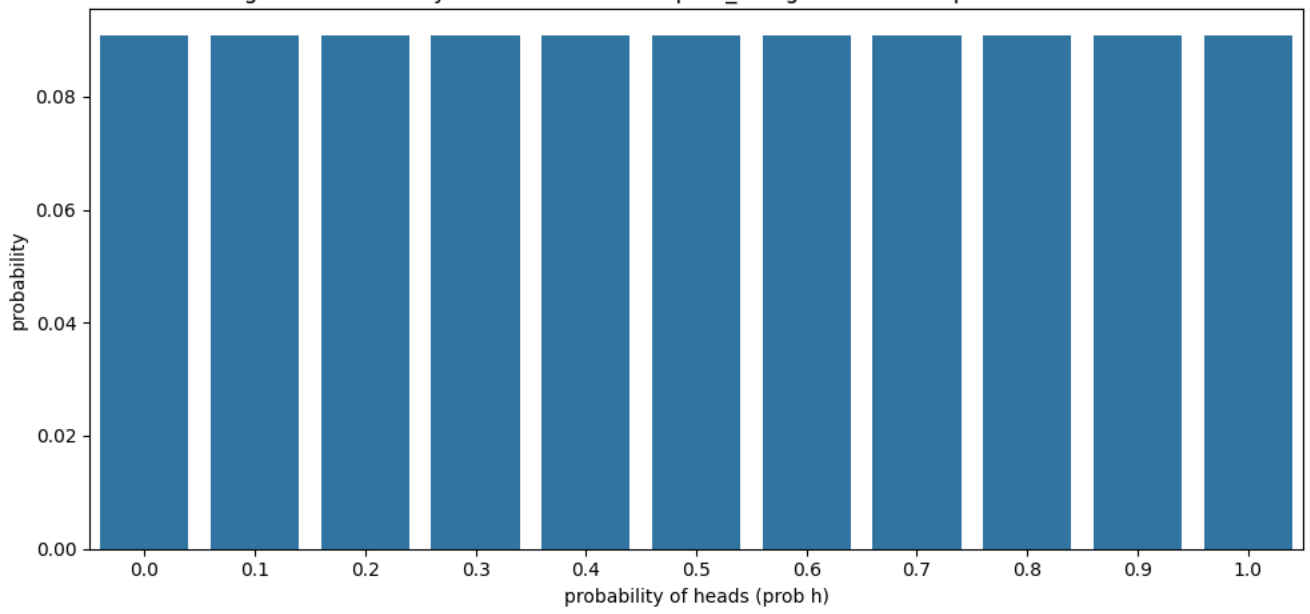
3. BAYESIAN PARAMETER ESTIMATION

For now, you have no information about the probability distribution the coin is coming from. So, you assume the probability of heads to be anywhere between 0 and 1 with each value being equally likely.

Each value can be seen as a hypothesis about the parameter value:

- $H_0: prob_{heads} = 0$
- $H_1: prob_{heads} = 0.1$
- $H_2: prob_{heads} = 0.2$
- $H_3: prob_{heads} = 0.3$
- $H_4: prob_{heads} = 0.4$
- …

Figure 2. Probability distribution of each prob_h in generated sample with mean 0.09

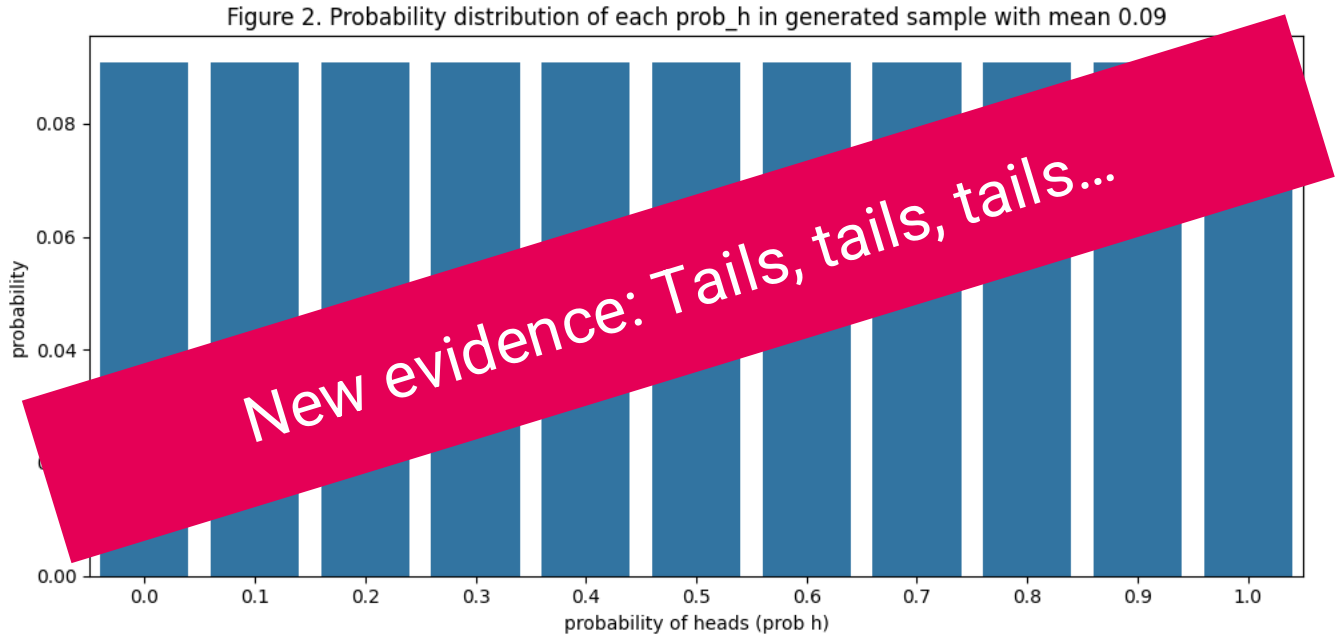


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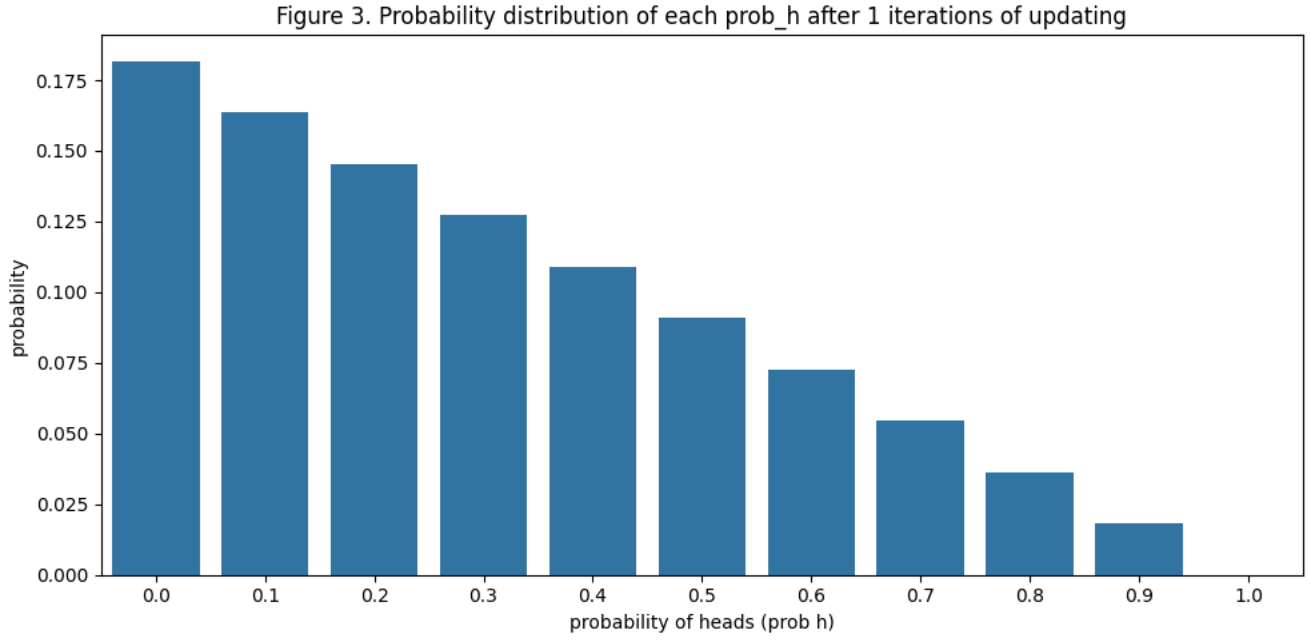
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- $H_2: prob_{heads} = 0.2$
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- $H_4: prob_{heads} = 0.4$
- …



3. BAYESIAN PARAMETER ESTIMATION

For now, you have no information about the probability distribution the coin is coming from. So, you assume the probability of heads to be anywhere between 0 and 1 with each value being equally likely.

$$p(prob_{heads} = 0.7|Tails) = \frac{0.3 * 0.09}{0.5} = 0.054$$



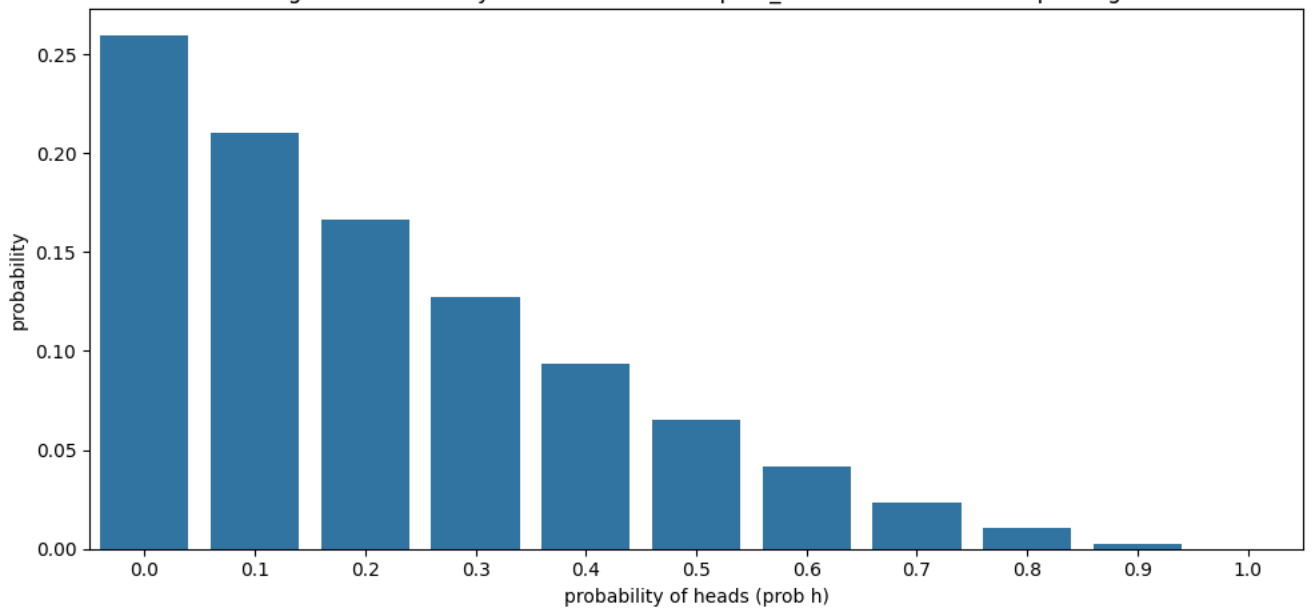
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$$p(prob_{heads} = 0.7|Tails) = \frac{0.3 * 0.09}{0.5} = 0.054$$

$$p(prob_{heads} = 0.7|Tails) = \frac{0.3 * 0.054}{0.5} = 0.032$$

Figure 3. Probability distribution of each prob_h after 2 iterations of updating



3. BAYESIAN PARAMETER ESTIMATION

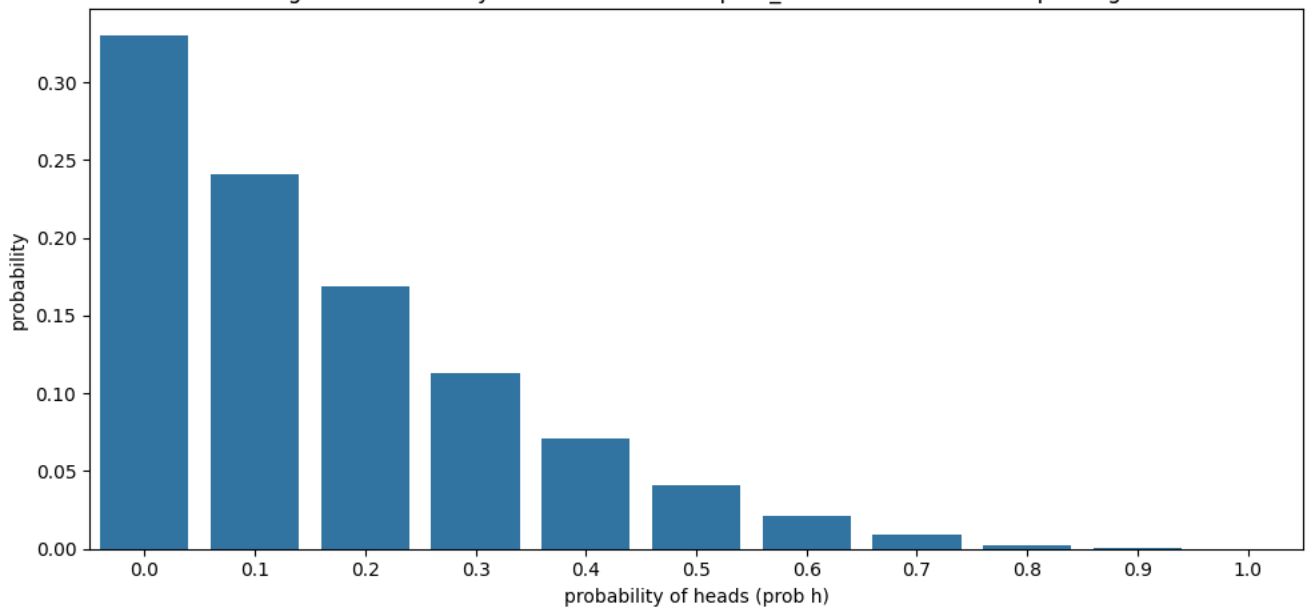
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$$p(prob_{heads} = 0.7|Tails) = \frac{0.3 * 0.054}{0.5} = 0.032$$

$$p(prob_{heads} = 0.7|Tails) = \frac{0.3 * 0.032}{0.5} = 0.019$$

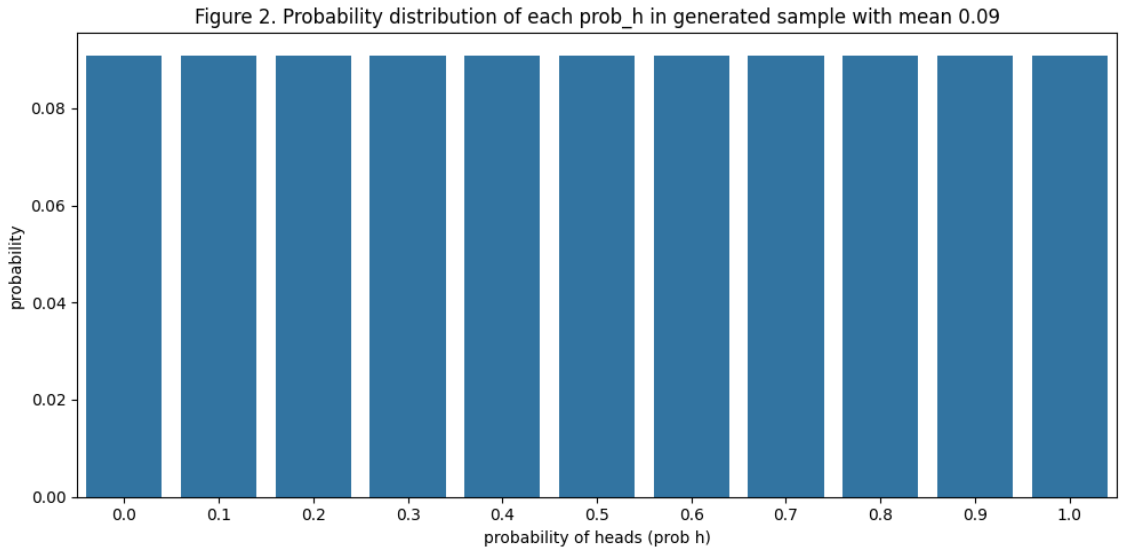
Figure 3. Probability distribution of each prob_h after 3 iterations of updating



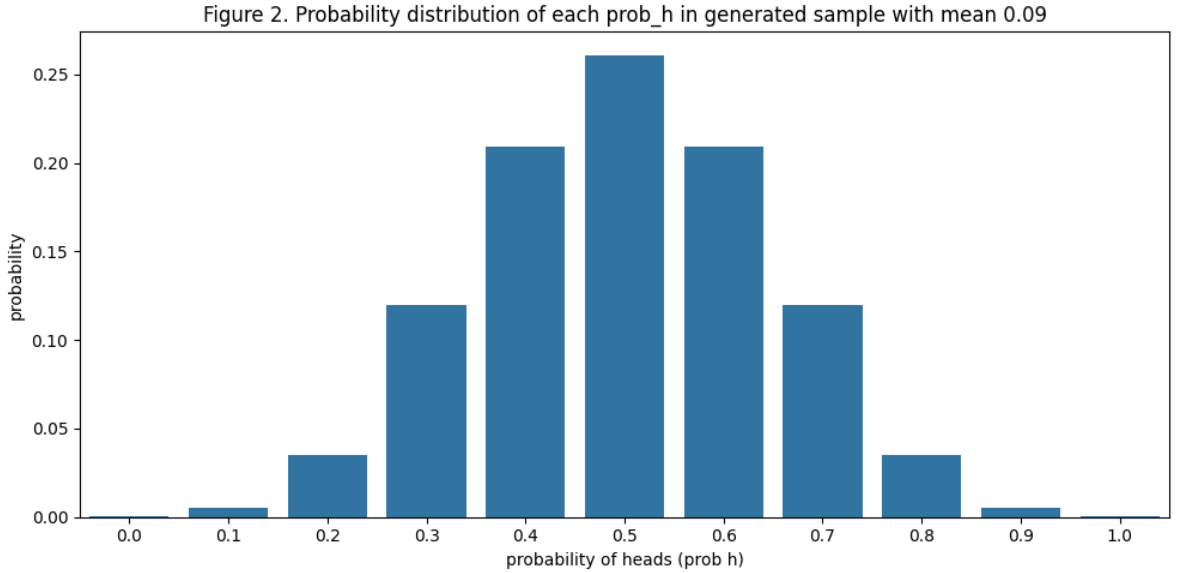
4. CHOOSING A PRIOR

So, how do you choose your prior?

Uniform prior



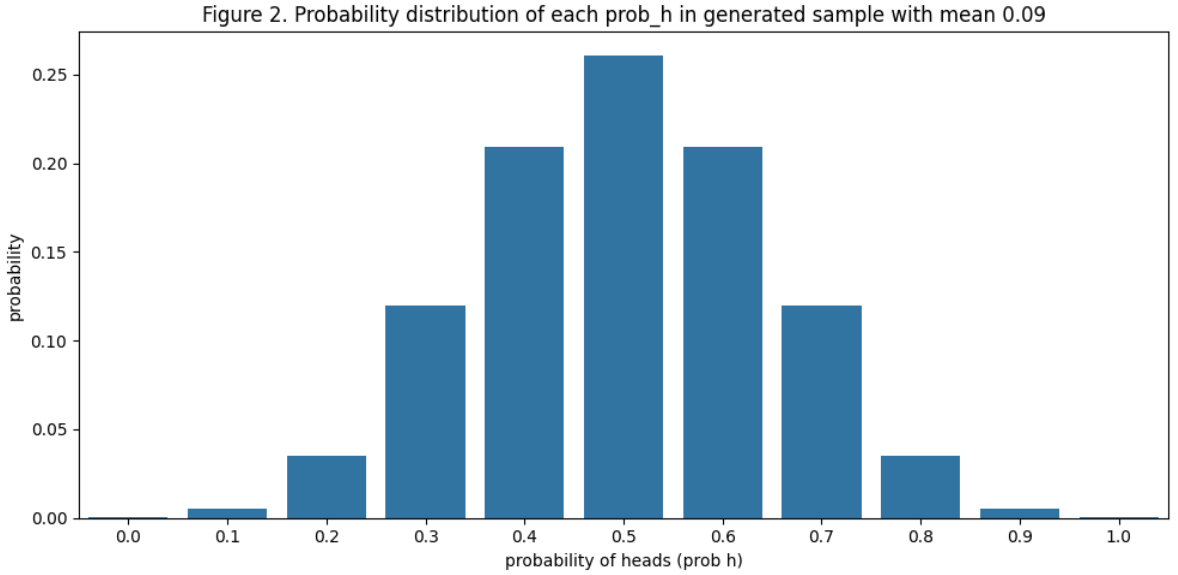
Gaussian prior



And all other possible distributions…

4. CHOOSING A PRIOR

From your previous knowledge you know most coins are fair or close to fair. So, you choose a gaussian prior that assumes most coins are fair.

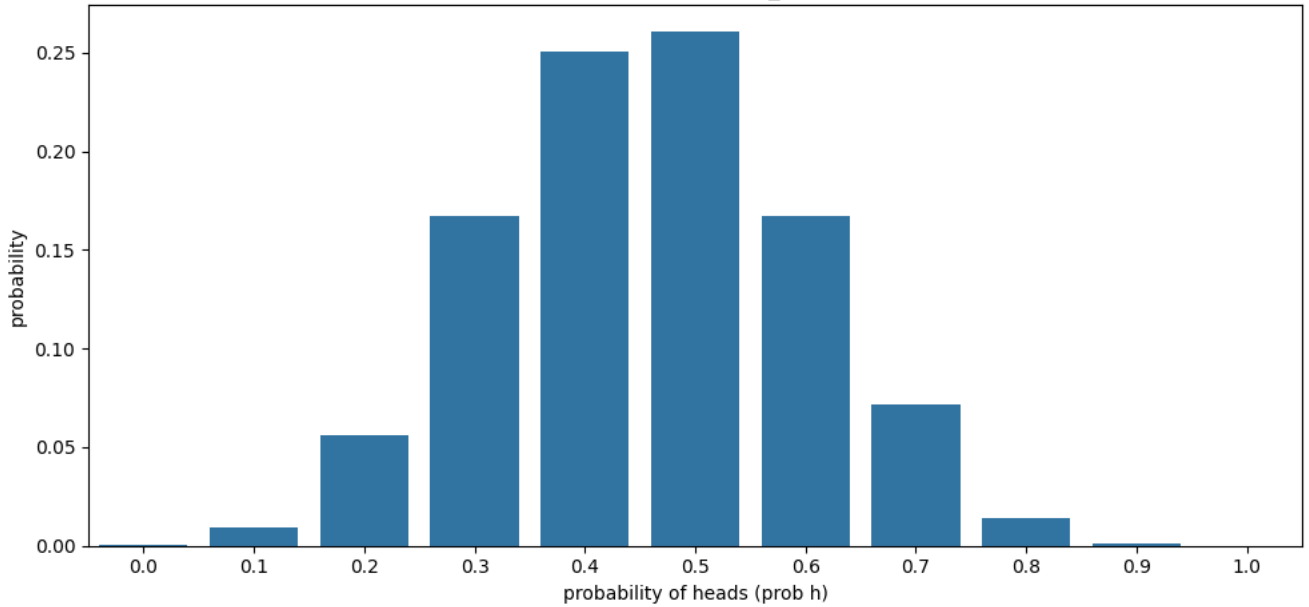


4. CHOOSING A PRIOR

From your previous knowledge you know most coins are fair or close to fair. So, you choose a gaussian prior that assumes most coins are fair.

$$p(prob_{heads} = 0.7|Tails) = \frac{0.3 * 0.12}{0.5} = 0.07$$

Figure 3. Probability distribution of each prob_h after 1 iterations of updating



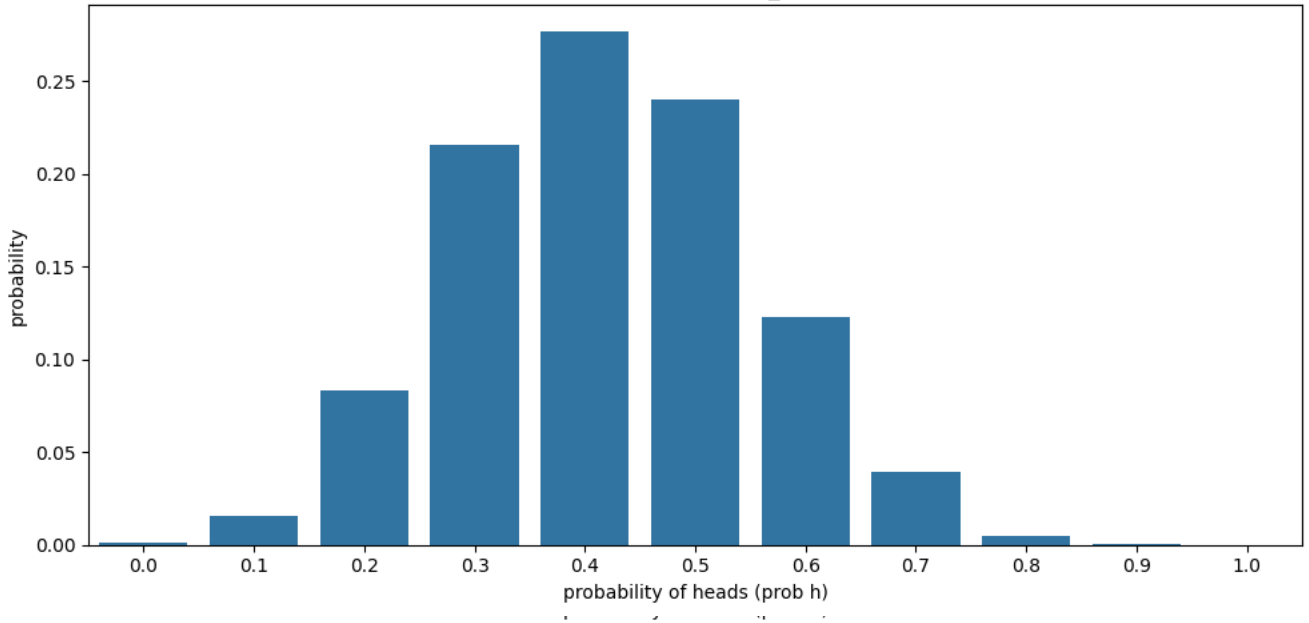
4. CHOOSING A PRIOR

From your previous knowledge you know most coins are fair or close to fair. So, you choose a gaussian prior that assumes most coins are fair.

$$p(prob_{heads} = 0.7|Tails) = \frac{0.3 * 0.12}{0.5} = 0.07$$

$$p(prob_{heads} = 0.7|Tails) = \frac{0.3 * 0.07}{0.5} = 0.04$$

Figure 3. Probability distribution of each prob_h after 2 iterations of updating



4. CHOOSING A PRIOR

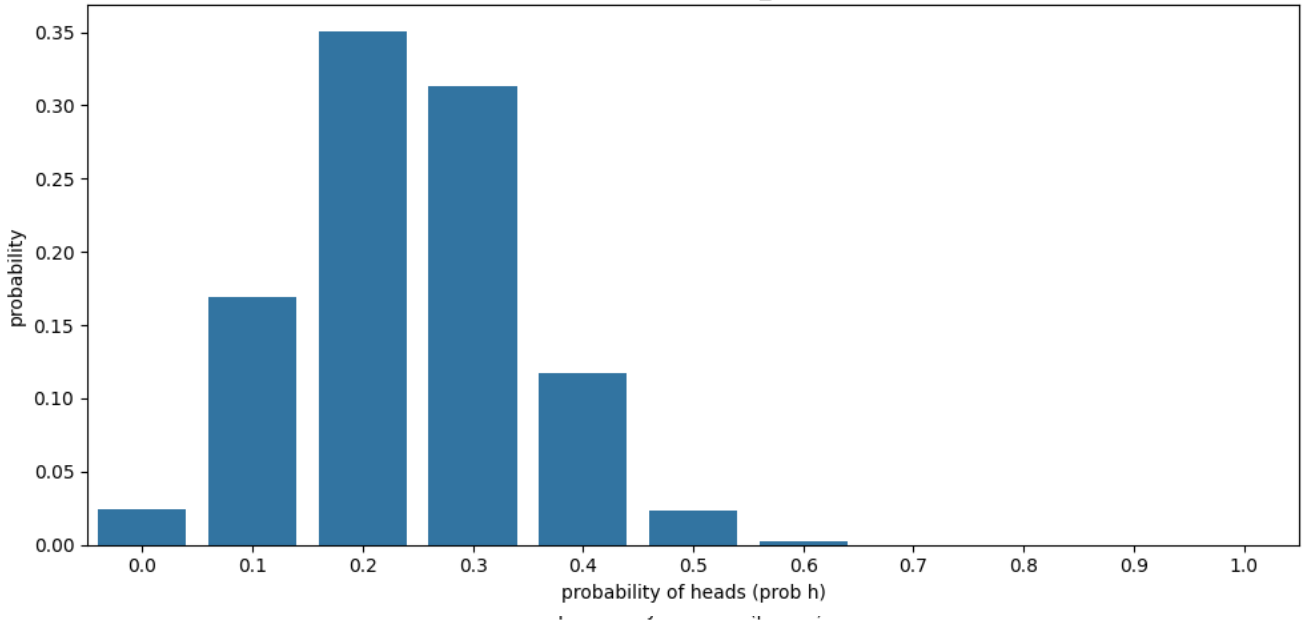
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$$p(prob_{heads} = 0.7|Tails) = \frac{0.3 * 0.12}{0.5} = 0.07$$

$$p(prob_{heads} = 0.7|Tails) = \frac{0.3 * 0.07}{0.5} = 0.04$$

$$p(prob_{heads} = 0.7|Tails) = 0.0007 \text{ (iteration 10)}$$

Figure 3. Probability distribution of each prob_h after 10 iterations of updating



4. CHOOSING A PRIOR

- The chosen prior distribution can have a **major impact** on the estimated posterior distribution
- Priors may be based on **historical data**, **expert beliefs** or a combination of both
- Choosing an **uninformative prior** skews results towards the data while choosing a **strong prior** skews results towards the prior
- Given enough data the estimated posterior will be attracted to the same distribution regardless of the prior (Bernstein-von Mises Theorem)

5. CONJUGATE PRIORS

- Computing the integral over each possible value of θ is expensive…

$$p(\theta|D) = \frac{p(D|\theta)p(\theta)}{p(D)} \qquad p(D) = \int p(D|\theta)p(\theta)d\theta$$

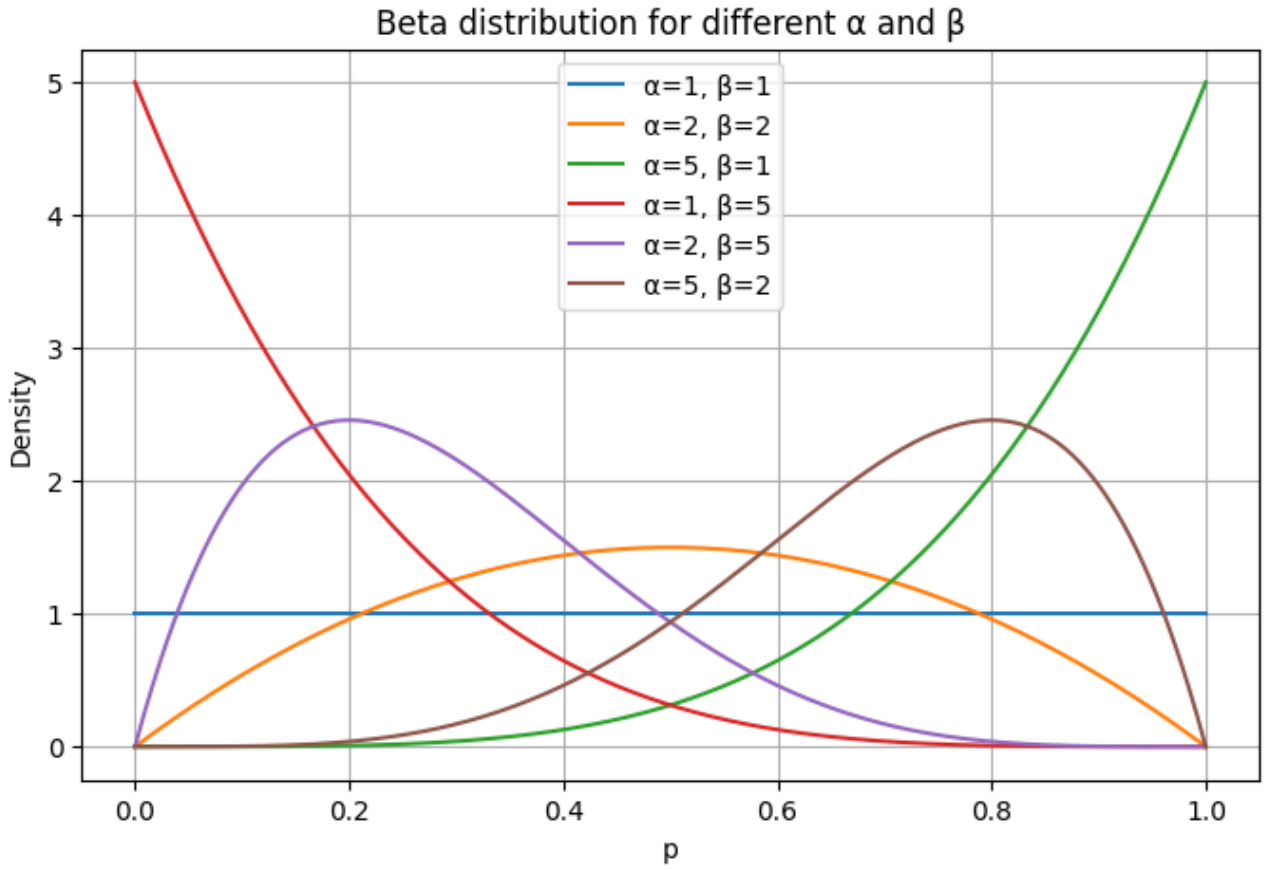
- Conjugate priors make updating priors mathematically convenient
- The posterior distribution has the same type of distribution as the prior

5. THE BETA DISTRIBUTION

- If we need probabilities of probabilities on binary outcomes the beta distribution provides a good approximation
  (ranges between 0 and 1)

- Updating simply comes down to counting

- It is a conjugate prior meaning that the posterior is again a beta distribution



5. THE BETA DISTRIBUTION

- The probability of a sequence of independent events is:

  $p(x_1)p(x_2), \ldots, p(x_n)$

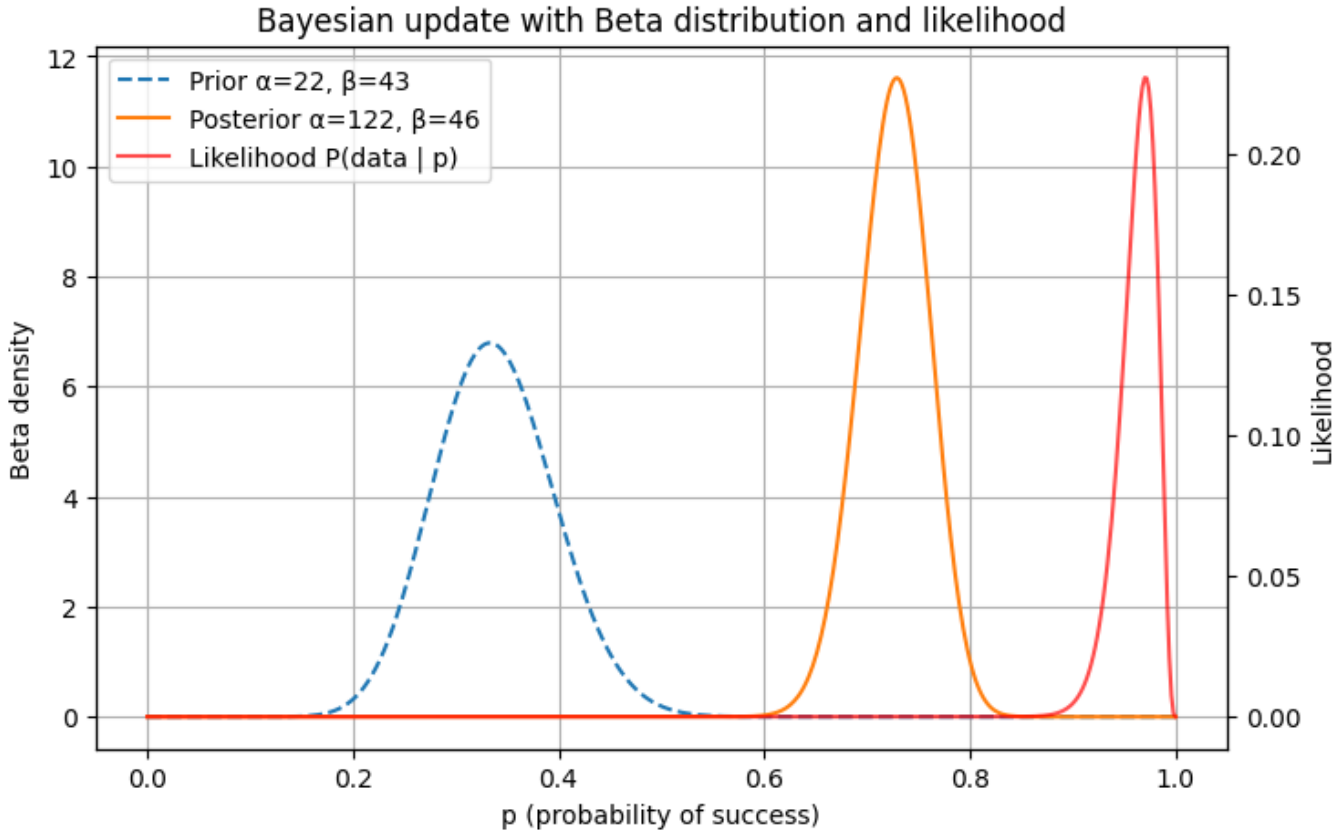
- For a binary sequence:

  $p(succes)^{\#successes}(1 - p(succes))^{\#failures}$

- Beta distribution with x probability of succes (your hypothesis), alpha the number of successes and beta the number of failures:

  $$\frac{1}{B(\alpha, \beta)} x^{\alpha-1}(1-x)^{\beta-1}$$

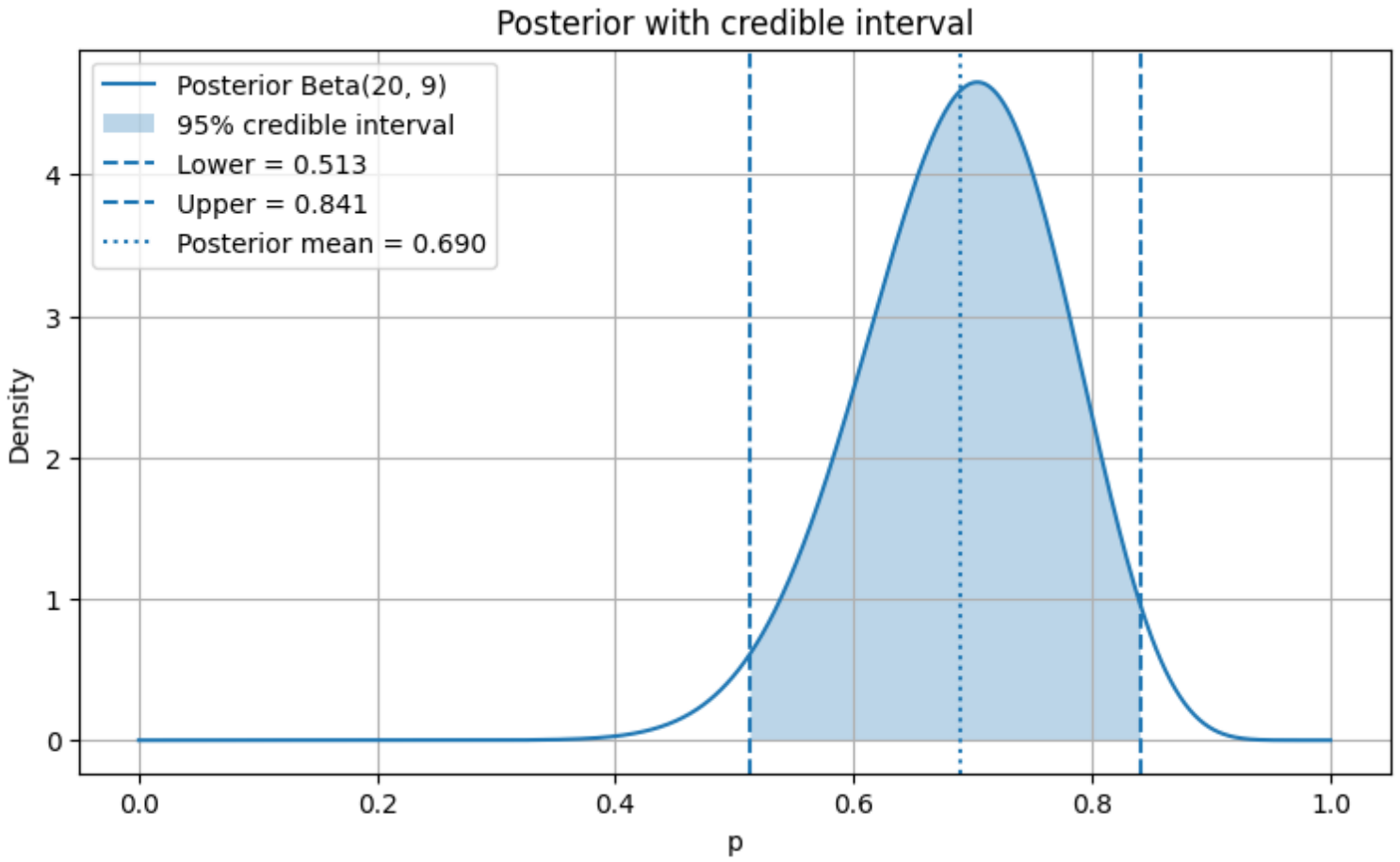
5. THE BETA DISTRIBUTION



5. CREDIBILITY INTERVALS

"There is a 95% probability that the true parameter value lies between [x,y] based on the data and the model."
(Credibility interval)

"If I take repeated random samples from the population in 95% of the time the population parameter falls into the range [x,y]."
(Confidence interval)



RECAP: THE BAYESIAN APPROACH TO STATISTICS

- Probability is viewed as a **subjective** experience of uncertainty; include **prior knowledge** in our judgements.
- Parameters are viewed as **probability distributions** that capture our uncertainty about their true value.
- **Continuously updating** what we believe about the world as **new information** comes in brings us closer to the truth.
- **Credibility intervals** reflect how uncertain our believes are.
- Bayesian theorem allows for **direct calculation of** the **probability** of our beliefs.

REACP: FREQUENTIST VS. BAYESIAN APPROACH

Frequentist approach

- Long-run frequency (repetition)
- Base results only on data
- Parameters are fixed single values
- Hypothesis testing (confidence intervals)

But,

- Probabilistic statements can only be made about the data
- Cannot incorporate additional knowledge
- Need to perform many experiments

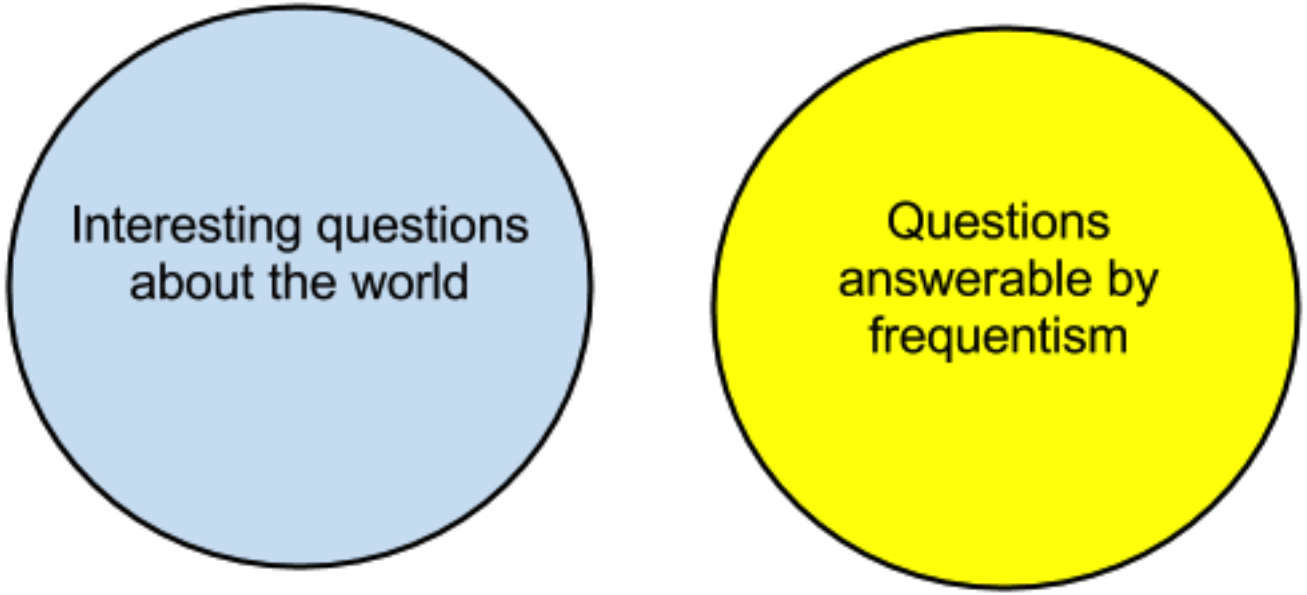
Bayesian approach

- Measures ‘degree of believe’
- Also include prior knowledge
- Parameters are probability distributions
- Updating priors (credibility intervals)

But,

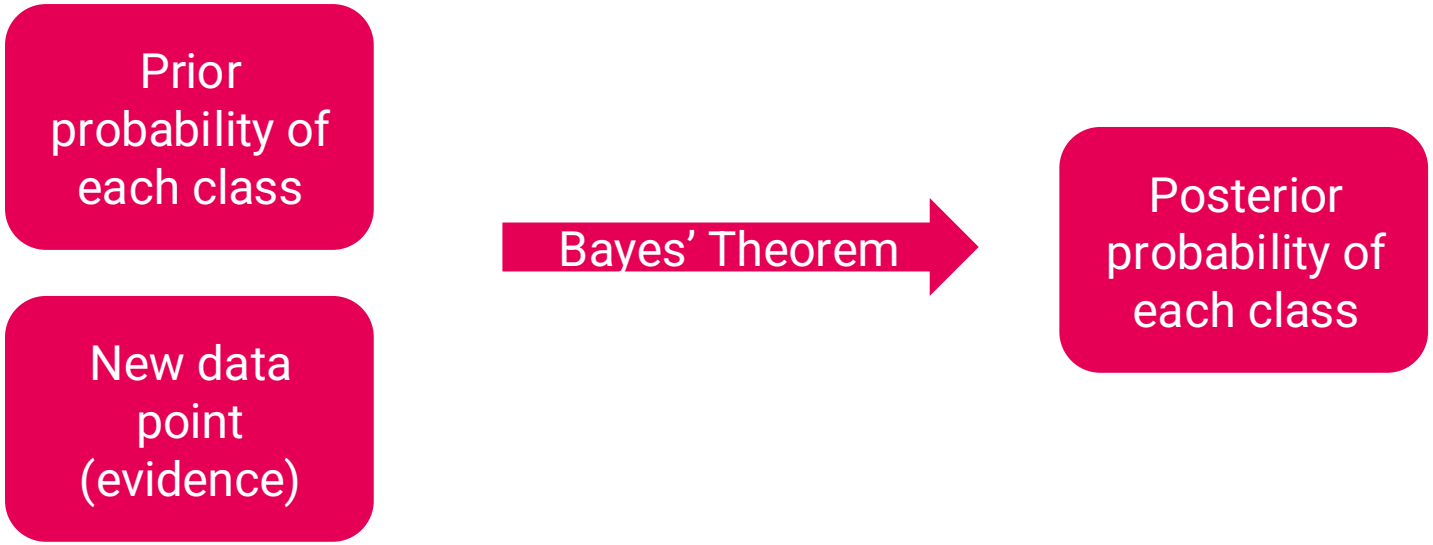
- Very sensitive to chosen prior
- Does not specify how to choose a prior (subjective)
- Greater technical complexity

5. FREQUENTIST VS. BAYESIAN APPROACH



6. NAÏVE BAYESIAN CLASSIFIERS

A **probabilistic classification** algorithm that predicts classes based on probability and uses **Bayes' theorem** to calculate the posterior probability of each class given the observed features.

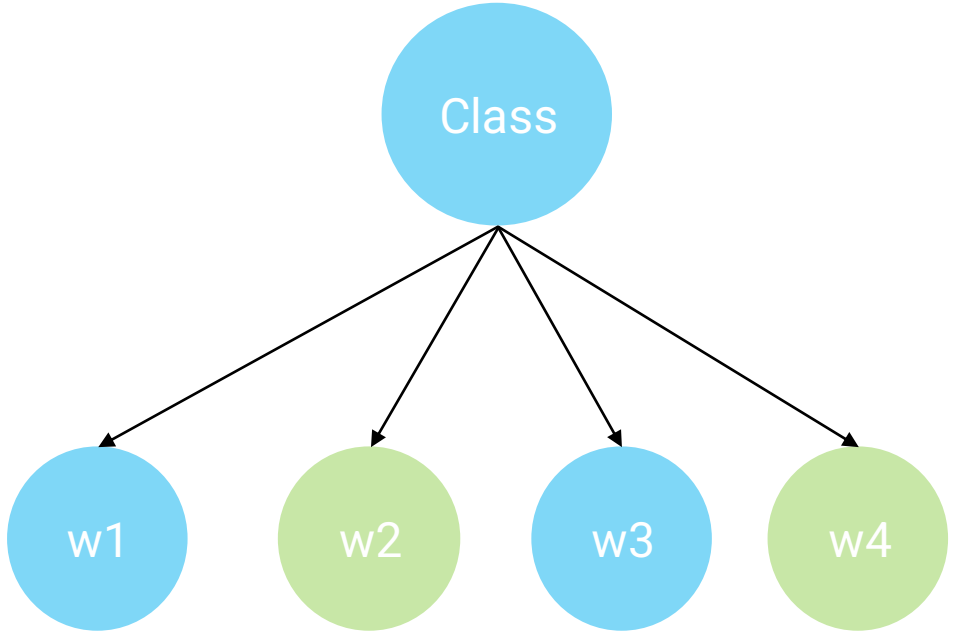


6. NAÏVE BAYESIAN CLASSIFIERS

Naïve Bayes for Spam filtering

- A set of mails with label $\{spam, normal\}$
- Use individual wordcounts as features
- Als $p(spam) > p(normal)$ classify mail as spam

Index	Word 1	Word 2	Word 3
Mail 1	3	2	0
Mail 2	3	0	8
Mail …			



6. NAÏVE BAYESIAN CLASSIFIERS

The Naïve Bayesian Classifier makes use of two simplifications:

1. Given the class label, the features (e.g., word occurrences) are assumed to be conditionally independent.

2. The model uses estimated probabilities (e.g., word frequencies within each class) rather than modelling full probability density functions.

6. NAÏVE BAYESIAN CLASSIFIERS

$$p(S|Content) = \frac{p(Content|S)p(S)}{p(Content)}$$

Basic Bayes Theorem

$$p(S|Content) = \frac{p(S|w_1)p(S|w_2)\dots p(S|w_n)}{p(S|w_1)p(S|w_2)\dots p(S|w_n) + (1-p(S|w_1))(1-p(S|w_2))\dots(1-p(S|w_n))}$$

Factorization of individual words

Divide by marginal to make sure we get a probability between 0 and 1.

$$p(S|Word) = \frac{p(W|S)p(S)}{p(W)}$$

Estimate 'spaminess' on word level and chain together to get spaminess of full content.

6. NAÏVE BAYESIAN CLASSIFIERS

Computing the conditional probability of spam given a specific word comes down to simple counting!

$$p(Spam|Word) = \frac{p(W|S)p(S)}{p(W)}$$

$$p(S) = \frac{\#spam\ messages}{\#all\ messages}$$

$$p(W) = \frac{\#all\ messages\ with\ word}{\#all\ messages}$$

$$p(W|S) = \frac{\#spam\ messages\ with\ word}{\#all\ spam\ messages}$$

6. NAÏVE BAYESIAN CLASSIFIERS

$p(S) = ?$

$p(Price) = ?$

$p(Price|S)$ = ?

$p(Hello) = ?$

$p(Hello|S)$ = ?

Index	Price	Won	Friend	Hello	Money
Spam	1	0	0	0	1
Normal	0	0	1	1	1
Normal	0	0	1	1	0
Spam	1	1	0	1	0
Spam	1	0	0	1	1
Normal	1	0	1	1	0

$$p(S) = \frac{\#spam\ messages}{\#all\ messages}$$

$$p(W) = \frac{\#all\ messages\ with\ word}{\#all\ messages}$$

$$p(W|S) = \frac{\#spam\ messages\ with\ word}{\#all\ spam\ messages}$$

6. NAÏVE BAYESIAN CLASSIFIERS

How to classify an email with words “Price” and ”Hello”?

$p(S) = \frac{3}{6} = 0.5$

$p(Price) = \frac{4}{6} = 0.66$

$p(Price|S) = \frac{3}{3} = 1$

$p(Hello) = \frac{5}{6} = 0.83$

$p(Hello|S) = \frac{2}{3} = 0.66$

$$p(Spam|Word) = \frac{p(W|S)p(S)}{p(W)}$$

$$p(Spam|Price) = \frac{1*0.5}{0.66} = 0.76$$

$$p(Spam|Hello) = \frac{0.66*0.5}{0.83} = 0.40$$

$$p(Spam|Content) = \frac{p(S|w_1) \dots p(S|w_n)}{p(S|w_1) \dots p(S|w_n) + (1 - p(S|w_1)) \dots (1 - p(S|w_n))}$$

$$p(Spam|Price, Hello) = \frac{0.76*0.40}{0.76*0.40+(1-0.76)(1-0.40)} = 0.68$$

6. NAÏVE BAYESIAN CLASSIFIERS

A few notes on stability:

1. Words that have not been seen in the training data have probability zero, we ignore these in the classification.

2. Factorization of probabilities can result in very small numbers fast; to prevent this Naïve Bayesian classifiers use logarithms of probabilities instead where small numbers become negatives instead of rounded to zero.

3. Very rare words can result in unreliable estimates. We use an alternative formula where probabilities are weighted with the frequency of words in the overall dataset.

6. NAÏVE BAYESIAN CLASSIFIERS

Advantages:

- Easy to implement
- Doesn’t require as much training data
- Handles continuous and discrete data
- Highly scalable and fast

Limitations:

- Strong (unrealistic) independence assumption
- Assigns zero probability to unseen cases
- Sensitive to overfitting

ASSIGNMENT

Walk through the naïve Bayesian Classifier example example for spam classification on Brightspace.

See if you can implement it with scikit-learn instead of manually